

# Nicholas W. Fraser

junior AI engineer · typescript · cloudflare · llm pipelines

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## 01 / SUMMARY

Junior AI engineer building production TypeScript and LLM applications on Cloudflare. Founder of **Tek Monkeys**, a Brisbane IT consultancy serving a paediatric orthopaedic practice, where I designed and deployed **Itemate**, an MBS billing validator pairing the Anthropic Claude API with a deterministic matcher and a server-side PHI-free data flow. Third-year UQ Business (Honours) student, Business Analytics + Finance double major, graduating 2027.

## 02 / EXPERIENCE

### Founder and AI Engineer @ Tek Monkeys 2024 - present

brisbane it consultancy · single-client paediatric orthopaedic practice · Brisbane, QLD

- Sole technical contact for a paediatric orthopaedic practice; manage IT support, hardware, and software within the practice's obligations under the Privacy Act 1988, the Australian Privacy Principles, and the My Health Records Act 2012.
- Hardware fleet includes a Windows server plus multiple clinical workstations (onsite and remote) and laptops.
- Scoped, designed, and shipped **Itemate**, the practice's MBS billing validator, after workflow interviews with the surgeons. Architecture detailed in [// projects](#).

### Hospitality & Customer Service @ Brisbane Airport 2023 - feb 2026

three years across four BNE food & beverage venues spanning the domestic and international terminals

- Customer-facing service in a high-volume transit retail environment: POS, cash and card handling, food prep, and rapid product knowledge for time-pressured passengers.
- Progressed from counter service to specialty coffee, finishing an eleven-month stint at Brisbane's most decorated specialty coffee brand (a former Australian Barista Champion's flagship).

### Dandelion & Driftwood specialty coffee · T1 international apr 2025 - feb 2026

Wolff Coffee Roasters' airport outpost; flagship owned by a former Australian Barista Champion and World Barista Championship judge.

### The Local restaurant & bar · T2 domestic jan - apr 2025

SSP's 425 m<sup>2</sup> flagship around a native fig tree; Queensland beer-garden concept showcasing local brewers and producers.

### Aviation Pier Cafe & Bar cafe & bar · T2 domestic 2024

Modern Australian runway-view cafe and bar championing Queensland producers (Green Beacon beer, Tamborine Mountain eggs).

### Mezze Za Za mediterranean counter · T2 domestic 2023 - 2024

Airport Retail Enterprises' chargrilled-souvlaki and mezze counter on the Qantas side of the domestic food court.

## A / SKILLS

### LANGUAGES

TypeScript · JS · Python · SQL · R

### AI ENGINEERING

Anthropic API · Claude Code · agent orchestration · hybrid LLM + deterministic · PHI pipelines

### FRAMEWORKS

SvelteKit 5 · React 19 · Vite · Bun · Vitest · R/Shiny · pandas · Chart.js · MapLibre

### PLATFORMS

Cloudflare Workers · Pages · Durable Objects · D1 · KV · Access · Drizzle · Web Crypto · GitHub Actions

### DATA & FINANCE

DCF + sensitivity modelling · derivatives pricing · interactive dashboards

### COMPLIANCE

Privacy Act 1988 + APPs · My Health Records Act 2012 · MBS / HIA 1973

## B / EDUCATION

### University of Queensland

B. Adv. Business (Hons), Business Analytics + Finance  
feb 2023 - dec 2027

BSAN2205 Machine Learning, BSAN3204 Data Viz, BSAN4204 Business Analytics, Corporate Finance, Asset Pricing, Derivatives.

## C / ADDITIONAL

### SOCIETIES

UQ Economics Society + Business Analytics Network Association (2023-).

### LANGUAGES

English (native) · Japanese (JLPT N4)

### VOLUNTEERING

Community BBQ for people experiencing homelessness (2020-2022).



SCAN FOR  
PORTFOLIO  
[nickwfraser.dev](https://nickwfraser.dev)

Itemate MBS billing validator

2026 - present

itemate.nickwfraser.dev

- TypeScript
- SvelteKit 5
- CF Pages + D1
- Drizzle
- Anthropic Claude API
- Vitest

- Built and deployed Itemate for a paediatric orthopaedic practice (two surgeons), pairing the Anthropic Claude API (Haiku for dev, Sonnet for prod) for clinical note normalisation with a deterministic keyword-overlap retrieval pass over a 6,039-item MBS index. Deployed at the edge behind Cloudflare Access (Google OAuth).
- Engineered a regex-anchored de-identification pipeline that strips PHI server-side before any LLM call; shipped 97 unit tests with GitHub Actions CI. No patient data persists on the server.

GoCard Insights QLD transit + mode-choice analytics

feb 2026

gocard.nickwfraser.dev

- TypeScript
- SvelteKit 5
- CF Pages + Workers
- D1 + KV
- Chart.js
- MapLibre GL

- Built and shipped the BSAN4204 Business Analytics research project, combining Go Card trip-history analysis with a live mode-choice calculator comparing public transport against driving costs after Translink's 50¢ flat-fare reform.
- Engineered a daily fuel-data pipeline as a sibling Cloudflare Worker cron against the QLD FPQ API with KV caching and D1 history retention; validated a pure mode-choice engine across 24 unit tests to within \$0.50 per week of manual spreadsheet calculations.

Monkey Barrel multiplayer web arcade

apr 2026 - present

arcade.nickwfraser.dev

- TypeScript
- SvelteKit 5
- CF Workers
- Durable Objects
- D1
- Web Crypto
- Bun

- Built and deployed a multiplayer party-game platform hosting 12+ games plus a 6,400-line idle-RPG sub-app (9 zones, 180 stages). Coordinated authoritative game state through Durable Objects on the WebSocket Hibernation API for near-zero idle cost; ~1,000 games played across 14 registered users since launch.
- Architected an actor-model room hierarchy (one Durable Object per room) with PBKDF2 + D1-backed session auth via Web Crypto; diagnosed and hotfixed a production incident where Workers' 100k PBKDF2 iteration cap blocked NIST G1 compliance.
- Caught a Durable Object hibernation regression across 5 room subclasses by running parallel Opus audit agents through Claude Code (23 findings with file-and-line precision in ~10 min); shipped fix as a unified connection model with 18 integration tests passing.